

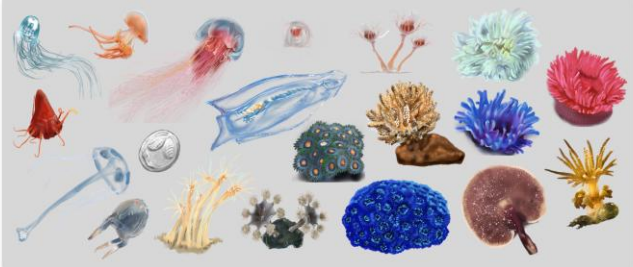
User manual

Home

CnidoSite

A Multi-Omics Database for Cnidarian Biological Study

Home Taxonomy Paleobiology Genome Transcriptome Single-cell Proteome Epigenome Metagenome Phenotype Tools Download Help



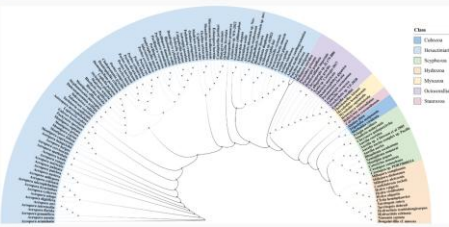
CnidoSite

The phylum Cnidaria represents a cornerstone of marine ecosystems and a pivotal lineage for understanding the evolution of multicellularity, tissue organization, and developmental complexity. Embracing corals, jellyfish, sea anemones, and hydroids, these organisms exhibit remarkable ecological diversity, dominating habitats ranging from vibrant tropical reefs to the vast expanse of the deep sea. Despite their ecological significance, cnidarian research has been constrained by fragmented, disparate omics datasets. To bridge this gap, we present **CnidoSite**, the first dedicated multi-omics atlas designed to systematically integrate functional genomic resources for cnidarian biological study.

CnidoSite provides an unprecedented scale of data coverage, integrating information from 7 classes, 22 orders, 107 families, and 325 species. The database houses 148 high-quality genome assemblies, 15,642 bulk transcriptomes, 33 single-cell datasets, 46 proteomic datasets, 723 epigenomic datasets, and 1,030 metagenomic samples. This is complemented by 151,046 phenotype records and 410 fossil entries. Functionally, the platform annotates 5,430,036 protein-coding genes, 98,834 transcription factors, 308,859 ubiquitin family members, and 3,621,064 gene families. Additionally, it incorporates 739,820,473 epigenetic modification peaks, 395 metagenome-assembled genomes (MAGs), and comprehensive annotations across nine major databases, including NR, UniProt, KEGG, and GO.

Beyond data aggregation, CnidoSite offers advanced analytical modules to decipher biological mechanisms. It features 31 co-expression networks comprising 13,793,211 co-expressed gene pairs to map regulatory landscapes across developmental stages. Leveraging 206 identified cell types and 297,645 cell markers, the database enables detailed cellular resolution of cnidarian biology. Furthermore, 398,247 microsatellite gene pairs are provided to elucidate pan-evolutionary genome structural features. As a centralized resource, CnidoSite serves as a vital platform for investigating larval settlement, symbiosis, and environmental adaptation, providing critical insights into the genetic underpinnings of marine biodiversity and evolution.

Cnidaria Phylogeny



Phylogenetic relationships of cnidarian species included in CnidoSite

Contact Us

For data inquiries, collaborations, or technical support, please contact our team:

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Learn more about our research: [Langshan Wu's Lab](#) | [Pei-Yuan Qian's Lab](#)

Data Statistics

Class/Order/Family/Species	7/22/107/325
Genomes/Datasets	76
Transcriptomic Datasets	15,642
Single-Cell Datasets	33
Proteomic Datasets	46
Epigenomic Datasets	723
Metagenomic Datasets	1,030
Phenotypic Datasets	151,046
Fossil Records	410
Protein-Coding Genes	5,430,036
Basic Genes	542,399
Co-Expressed Networks	31
Co-Expressed Gene Pairs	13,793,211
Cell Types	206
Cell Markers	297,645
Quantified Proteins	83,306
Epigenetic Modification Type	13
Epigenetic Modification Peaks	739,820,473
Metagenome-Assembled Genomes	395
Type of Gene Annotation	9
NR Database	95,362,542
UniProt Database	140,048,912
KEGG Pathways	1,362,360
Transcription Factors	98,834
Ubiquitin Family	308,859
Gene Ontology	36,871,005
Interpro Annotation	35,694,300
Plant Domain	7,475,362
RAxML	3,882,015
Gene Family	3,621,064

External Resources

- The Cnidarian Evolutionary Genomics Database
- Cnidarians - Animal Database
- World Register of Marine Species (WoRMS)
- Ocean Biodiversity Information System (OBIS)
- Global Biodiversity Information Facility (GBIF)
- The Paleobiology Database (PBDB)

Recent Updates

- 2020-05-26** Pan-genome and synteny analysis enabled
- 2020-05-14** Phenotypic datasets enabled
- 2020-05-10** Single-cell omics pages are available
- 2020-04-07** Functional enrichment analysis module launched
- 2020-04-05** BLAST search tool integrated
- 2020-04-04** Taxonomic browsing fully implemented
- 2020-03-02** Gene co-expression networks available
- 2020-02-19** Primer design tool added
- 2020-01-29** Browse genome visualization deployed
- 2020-12-28** Fossil record integration completed
- 2020-11-18** CnidoSite initial release

Detailed introduction to the CnidoSite database

Summary of data composition in CnidoSite database

Tree phylogeny for 148 Cnidarian species

Associated weblinks

News and updates

Taxonomy

Taxonomy Overview

In this module, the taxonomy coverage of 325 Cnidarian organisms is listed.

Phylum	Class	Order	Family	Genus	Species	NCBI Taxonomy ID	WoRMS ID	GBIF ID
Cnidaria	Cubozoa	Carybdeida	Alatinidae	unclassified Alatinidae	Alatinidae sp. Z8VKAUB7J3	2292721	-	-
Cnidaria	Cubozoa	Carybdeida	Alatinidae	Alatina	Alatina alata	1193083	289378	2264702
Cnidaria	Cubozoa	Chirodropida	Chirodropidae	Chironex	Chironex yamaguchii	672230	395227	4340380
Cnidaria	Cubozoa	Carybdeida	Tripedaliidae	Tripedalia	Tripedalia maipoensis	2949330	1649593	12185925
Cnidaria	Cubozoa	Carybdeida	Carybdeidae	Carybdea	Carybdea cf. marsupialis auct. non (Linnaeus, 1758)	686687	-	-
Cnidaria	Cubozoa	Carybdeida	Tamoyidae	Tamoya	Tamoya ohboya	685043	559023	5842335
Cnidaria	Cubozoa	Carybdeida	Carukiidae	Morbakka	Morbakka virulenta	686327	484726	5842338
Cnidaria	Hexacorallia	Scleractinia	Oculinidae	Madrepora	Madrepora sp. 1 SH-2022	2982801	-	-
Cnidaria	Hexacorallia	Scleractinia	Oculinidae	Madrepora	Madrepora sp. SY280-shi	3117734	-	-
Cnidaria	Hexacorallia	Scleractinia	Mussidae	Pseudodiploria	Pseudodiploria clivosa	1917265	828326	8377035



This module provides taxonomy overview for 325 cnidarian species. Users can click links with blue color for details

 Total Taxon Information: 325

Show:

10 / page

 records per page

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Paleobiology

Paleobiology

In this module, a brief summary of **410 fossil records** for each taxon of Cnidaria species collected from [the Paleobiology Database \(PBDB\)](#) is provided.

Phylum	Class	Order	Family	Genus	Species
Cnidaria	Octocorallia	Malacalcyonacea	Tubiporidae	Tubipora	Tubipora musica
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora abrotanoides
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora acuminata
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora aspera
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora austera
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora branchi
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora cerealis
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora cervicornis
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora clathrata
Cnidaria	Hexacorallia	Scleractinia	Acroporidae	Acropora	Acropora cytherea



This module provides 410 fossil records for cnidarian species collected from the Paleobiology Database (PBDB). Users can click links with blue color for details.

Total Fossil Records: 410

Show: 10 /page records per page

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Genomic Data

Species Overview

In this module, a brief summary of 325 assembled Cnidarian genomes is provided. In addition, the detailed information for genomes and relevant literatures are also provided.

Class	Species Latin Name	NCBI Taxonomy ID	Size (Mb)	Assembl Level	Released Year	Pubmed ID
Cubozoa	Alatina alata	1193083	1247.88	Scaffold	2022	31257419
Cubozoa	Alatinidae sp. Z8VKAUB7J3	2292721	1262.41	Scaffold	2023	-
Cubozoa	Carybdea cf. marsupialis auct. non (Linnaeus, 1758)	686687	1162.16	Scaffold	2023	-
Cubozoa	Chironex yamaguchi	672230	445.68	Scaffold	2022	-
Cubozoa	Morbakka virulenta	686327	907.49	Scaffold	2019	31048744
Cubozoa	Tamoya ohboya	685043	175.43	Scaffold	2023	-
Cubozoa	Tripedalia malpoensis	2949330	608.25	Chromosome	2026	41877162
Hexacorallia	Acropora abrotanoides	676057	408.36	Scaffold	2026	-
Hexacorallia	Acropora acuminata	367763	376.45	Scaffold	2020	34151991
Hexacorallia	Acropora austera	117779	459.85	Chromosome	2024	42218227
Hexacorallia	Acropora awl	1929007	408.98	Scaffold	2020	34151991
Hexacorallia	Acropora cervicornis	6130	293.2	Scaffold	2023	37804092
Hexacorallia	Acropora cf. manni	3471590	437.99	Chromosome	2025	42218227
Hexacorallia	Acropora cytherea	46686	406.58	Scaffold	2020	34151991
Hexacorallia	Acropora digitifera	70779	396.56	Scaffold	2020	34151991
Hexacorallia	Acropora echinata	513064	382.88	Scaffold	2020	34151991
Hexacorallia	Acropora florida	70782	422.31	Scaffold	2020	34151991
Hexacorallia	Acropora gemmifera	70780	382.42	Scaffold	2020	34151991

→ This module offers detailed information, including genome assembly, BUSCO completeness, gene number and corresponding published literatures.

Gene Search

Gene detailed information

Detailed information of XP_032233839.1 in *Nematostella vectensis*

Genomic Location: NC_064035.1:18240001...18240008

Ref: accession: XP_032233839.1, ATP-binding cassette sub-family F member 2 (Nematostella vectensis)

Gene Structure

Exons Intron

[More details in Jbrowse](#)

Sequence

CD1

Transcript

Protein

Uniprot

Uniprot name	Description
Q95833	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)
Q95834	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)
Q95835	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)

Gene family

Gene family	Subfamily

Pfam domain

Pfam accession	Pfam name	Description	Type	Source
PF00067	ABC_1	ABC transporter	C domain	InterPro
PF00068	ABC_2	ABC transporter	C domain	InterPro

InterPro

InterPro name	Type	Description	Source
IPR000603	Domain	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro
IPR000604	Domain	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro
IPR000605	Family	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro
IPR000606	Family	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro
IPR000607	Domain	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro

PANTHER

PANTHER name	Description	Source
P000001	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro

Gene Ontology

GO term	Category	Description	Source
GO:0005575	Molecular Function	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro
GO:0005576	Molecular Function	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	InterPro

KEGG pathway

KEGG ID	Enzyme	Enzyme ID	Pathway	Enzyme	Source
K04463	ATP-binding cassette sub-family F member 2 (Nematostella vectensis)	-	Glucose degradation	K04463	InterPro

Search

Search genes, genomic loci, and functional annotations across Cnidaria species. Explore detailed gene information, protein domains, and biological functions.

Gene & Locus Search

Gene ID Search



Class

Hexactinaria

Species

Nematostella vectensis

Genomic Locus Search



Class

Hexactinaria

Species

Nematostella vectensis

Gene ID

Enter gene identifier

Example: XP_032233839.1

Genomic Position

Enter genomic coordinates

Example: NC_064035.1:18240000-18240000

Search Gene

Reset Form

Search Locus

Reset Form

1. Gene ID search

2. Genomic region search

Genes related to the genomic region

Locus Search Results

We find 7 records based on your search.

Species	Gene	Annotation	Chr	Start	End	Strand	Jbrowse
Nematostella vectensis	XP_032233841.2	YLP MOTIF CONTAINING PROTEIN NUCLEAR PROTEIN ZAP	NC_064035.1	18230429	18240802	-	Jbrowse
Nematostella vectensis	XP_032233839.1	ATP-BINDING TRANSPORT PROTEIN-RELATED	NC_064035.1	18240851	18248028	+	Jbrowse
Nematostella vectensis	XP_048579479.1	-	NC_064035.1	18248392	18250621	-	Jbrowse
Nematostella vectensis	XP_032233844.2	THUMP DOMAIN CONTAINING PROTEIN 1-RELATED	NC_064035.1	18250741	18258820	+	Jbrowse
Nematostella vectensis	XP_032233846.2	THUMP DOMAIN CONTAINING PROTEIN 1-RELATED	NC_064035.1	18250746	18258820	+	Jbrowse
Nematostella vectensis	XP_032233842.1	CELL DIVISION PROTEIN KINASE	NC_064035.1	18257639	18262987	-	Jbrowse

Gene Search

 Functional Annotation Search

Keyword Search

Class

Hexactiniaria

Species

Nematostella vectensis

Biological Keywords

Enter functional keywords

Example: ABC transporter

Search Function

Reset Form

Protein Domain Search

Class

Hexactiniaria

Species

Nematostella vectensis

Protein Domain ID

Enter Pfam domain identifier

Example: PF12848

Search Domain

Reset Form

3. Gene function search



Genes related to the provided keywords

Keywords Search Results

Detailed information based on your search.

Total Search Records in *Nematostella vectensis*: 1499

Show: 10/page genes per page

Species	Gene	InterPro ID	Annotation	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR017871	ABC transporter-like, conserved site	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR011527	ABC transporter type 1, transmembrane domain	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR011527	ABC transporter type 1, transmembrane domain	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR003439	ABC transporter-like, ATP-binding domain	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR011527	ABC transporter type 1, transmembrane domain	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR017871	ABC transporter-like, conserved site	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR003439	ABC transporter-like, ATP-binding domain	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR036640	ABC transporter type 1, transmembrane domain superfamily	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR003439	ABC transporter-like, ATP-binding domain	Jbrowse
Nematostella vectensis	XP_048589316.1	IPR036640	ABC transporter type 1, transmembrane domain superfamily	Jbrowse

4. Protein domain search



Genes related to the provided protein domain

Pfam Search Results

Detailed information based on your search.

Total Search Records in *Nematostella vectensis*: 3

Show: 20/page genes per page

Species	Gene	Pfam accession	Pfam name	Description	Type	Jbrowse
Nematostella vectensis	XP_048587329.1	PF12848	ABC_tran_Xtn	ABC transporter	Domain	Jbrowse
Nematostella vectensis	XP_032232368.1	PF12848	ABC_tran_Xtn	ABC transporter	Domain	Jbrowse
Nematostella vectensis	XP_032233839.1	PF12848	ABC_tran_Xtn	ABC transporter	Domain	Jbrowse

Benchmarking Universal Single-Copy Ortholog (BUSCO) genes were identified for all cnidaria species. You can select different class or species to see details such as the busco id, gene id, status and length. Detailed gene structure and sequences are shown in JBrowse.

Select Species for BUSCO Analysis

Select Class

Hexactinaria

Select Species

Acropora acuminata

View BUSCO Details

Reset Selection



BUSCO Gene Details

Detailed Benchmarking Universal Single-Copy Ortholog (BUSCO) analysis results for the selected species.

Total BUSCO Genes in *Acropora acuminata*: 3466

Show: 10 /page genes per page

Busco ID	Status	Score	Length	Gene ID	Description	Visualization
63at6073	Duplicated	834.2	365.0	aacu_s0202.g18.t2	testin-like	JBrowse
63at6073	Duplicated	860.1	362.0	aacu_s0202.g18.t1	testin-like	JBrowse
65at6073	Complete	472.1	232.0	aacu_s0012.g141.t1	MORN repeat-containing protein 3	JBrowse
84at6073	Complete	349.8	157.0	aacu_s0061.g5.t1	SUMO-conjugating enzyme UBC9	JBrowse
149at6073	Complete	475.0	216.0	aacu_s0163.g50.t1	cytochrome b5 domain-containing protein 1	JBrowse
207at6073	Complete	606.5	333.0	aacu_s0258.g23.t1	fatty acid 2-hydroxylase	JBrowse
227at6073	Complete	688.4	313.0	aacu_s0005.g358.t1	guanine nucleotide-binding protein subunit beta-2-like 1	JBrowse
232at6073	Complete	735.3	403.0	aacu_s0073.g82.t1	zinc finger protein ZIC 4	JBrowse
233at6073	Complete	332.2	218.0	aacu_s0030.g88.t1	uncharacterized protein LOC116298145	JBrowse
302at6073	Complete	579.6	317.0	aacu_s0050.g91.t1	vesicular integral-membrane protein VIP36-like	JBrowse

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Detailed information including busco id, status, score, length, gene and Jbrowse visualization

Transposable Elements

Transposable Elements

Several studies have shown that transposable elements (TEs) play a crucial role in genome evolution and adaptation, particularly in deep-sea animals. TEs can drive genomic innovation by creating genetic diversity, facilitating gene duplication, and promoting genome rearrangements. This module provides the information of detailed TEs correlated with protein-coding genes for subsequent functional analysis. Users can search TEs by **gene name**, **TE subfamily type**, **genomic regions** and additional links to Jbrowse are provided for details.

 Search TEs by Gene or TE Type

Class:

Species:

Gene/TE Type:

 Example: Actl_000001-T1

1. Gene ID or TE type search



TE locus/type, related gene and distribution

 Search Parameters

SPECIES <i>Actinermus sp. WN-2022</i>	SEARCH TERM Actl_000001-T1	TOTAL RESULTS 115 TE(s) found
--	-------------------------------	----------------------------------

Species	TE ID	Scaffold	Start	End	Related Gene	Distribution	TE Type	Jbrowse
Actinermus sp. WN-2022	TE_homo_0	SCLC4GM_1	68	199	Actl_000001-T1	intergenic region	CACTA_TIR_transposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_1	SCLC4GM_1	234	334	Actl_000001-T1	intergenic region	LTR_retrotransposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_2	SCLC4GM_1	668	763	Actl_000001-T1	intergenic region	L2_LINE_retrotransposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_3	SCLC4GM_1	1334	1423	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_4	SCLC4GM_1	1429	1522	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_5	SCLC4GM_1	1536	1624	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_6	SCLC4GM_1	1629	1723	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_7	SCLC4GM_1	1730	1835	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_8	SCLC4GM_1	2122	2204	Actl_000001-T1	intergenic region	LTR_retrotransposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_9	SCLC4GM_1	2179	2266	Actl_000001-T1	intergenic region	Mutator_TIR_transposon	Jbrowse

 Transposable Elements Related to "Actl_000001-T1": 115 Records

Show: records per page

 Search TEs by Genomic Region

Class:

Species:

Genome Region:

 Example: SCLC4GM_1:1000-10000000

2. Genomic region search



Genes related to the genomic region

 Search Parameters

SPECIES <i>Actinermus sp. WN-2022</i>	GENOMIC REGION SCLC4GM_1:1000-10000000	TOTAL RESULTS 350 TE(s) found
--	---	----------------------------------

Species	TE ID	Scaffold	Start	End	Related Gene	Distribution	TE Type	Jbrowse
Actinermus sp. WN-2022	TE_homo_3	SCLC4GM_1	1334	1423	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_4	SCLC4GM_1	1429	1522	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_5	SCLC4GM_1	1536	1624	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_6	SCLC4GM_1	1629	1723	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_7	SCLC4GM_1	1730	1835	Actl_000001-T1	intergenic region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_8	SCLC4GM_1	2122	2204	Actl_000001-T1	intergenic region	LTR_retrotransposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_9	SCLC4GM_1	2179	2266	Actl_000001-T1	intergenic region	Mutator_TIR_transposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_10	SCLC4GM_1	2230	2352	Actl_000001-T1	gene region	helitron	Jbrowse
Actinermus sp. WN-2022	TE_homo_11	SCLC4GM_1	2588	2853	Actl_000001-T1	gene region	CACTA_TIR_transposon	Jbrowse
Actinermus sp. WN-2022	TE_homo_12	SCLC4GM_1	2963	3340	Actl_000001-T1	gene region	helitron	Jbrowse

 Transposable Elements Related to "SCLC4GM_1:1000-10000000": 350 Records

Show: records per page

TFs/Ubs

Select Gene Family Parameters

Class:

Species:

Gene Family:

[View Gene Family Members](#) [Reset Form](#)

1. Gene family search

2. Statistics of different types of gene family

Transcription Factor Families				
z1-C2H2(156)	TF_bZIP(8)	PC4(16)	z1-BED(26)	C3D2(2)
MYB(5)	Homeobox(36)	HTH(9)	ETS(4)	ARID(5)
bHLH(38)	THAP(75)	z1-GAGA(3)	Fork_head(14)	ZBTB(18)
MH(4)	DM(2)	PS(2)	AP-2(3)	HMG(13)
T-box(7)	CS(1)	CUT(1)	GCNF-like(3)	z1-CCCH(3)
ESR-like(2)	Miscellaneous(2)	NGF8-like(2)	RXR-like(2)	SF-like(1)
Pou(2)	RHD(1)	PRX(7)	Run1(1)	CBF(2)
HSF(1)	TEA(1)	z1-C2HC(1)	THR-like(1)	z1-NF-X(1)
GCF(1)				

3. Family member information of specific type of gene family

>> Gene Family>> TF family>> TF_bZIP

Gene	Plasm Self-build	DNA Binding Domain	Full name	Description	Jbrowse
Aiala_g1014.11	self-build	bZIP	TF_bZIP	dipeptidyl peptidase 2 [Hydra vulgaris]	Jbrowse
Aiala_g1049.11	self-build	bZIP	TF_bZIP	PREDICTED: zinc finger protein ZPR1 [Stomoxys calcitrans]	Jbrowse
Aiala_g1534.11	self-build	bZIP	TF_bZIP	LOW QUALITY PROTEIN: uncharacterized protein LOC110243672 [Ekaipasia diaphana]	Jbrowse
Aiala_g1685.11	self-build	bZIP	TF_bZIP	Retrovirus-related Pol polyprotein from transposon TNT 1-94 [Styloghara pistillata]	Jbrowse
Aiala_g1779.11	self-build	bZIP	TF_bZIP	unnamed protein product [Rotaria socialis]	Jbrowse
Aiala_g1805.11	self-build	bZIP	TF_bZIP	uncharacterized protein LOC124456927 [Xenia sp. Carnegie-2017]	Jbrowse
Aiala_g2010.11	self-build	bZIP	TF_bZIP	Retrovirus-related Pol poly from transposon [Paramuricea clavata]	Jbrowse
Aiala_g441.11	self-build	bZIP	TF_bZIP	Craniotfacial development protein 2 [Lamellibrachia satsuma]	Jbrowse

Protein Domain

 Input Gene List to Browse Protein Domain

Select Class

Hexactiniaria

Select Species

Nematostella vectensis

Gene List

XP_032229903.1
XP_048584816.1
XP_032232711.2
XP_001638735.3
XP_001630088.2
XP_048576726.1
XP_032232711.2

Example: Nematostella vectensis

Submit Query

Reset Selection

Gene list search



Protein domain information for the gene list

Total Protein Domains for 20 Genes: 35

Show: 10 /page pathways per page

Gene	Pfam accession	Pfam name	Description	Type	JBrowse
XP_032229903.1	PF13855	LRR_8	Leucine rich repeat	Repeat	JBrowse
XP_032229903.1	PF00560	LRR_1	Leucine Rich Repeat	Repeat	JBrowse
XP_048584816.1	PF00071	Ras	Ras family	Domain	JBrowse
XP_032232711.2	PF00071	Ras	Ras family	Domain	JBrowse
XP_001638735.3	PF00001	7tm_1	7 transmembrane receptor (rhodopsin family)	Family	JBrowse
XP_001630088.2	PF01019	G_glu_transpept	Gamma-glutamyltranspeptidase	Domain	JBrowse
XP_048576726.1	PF00008	EGF	EGF-like domain	Domain	JBrowse
XP_048576726.1	PF02210	Laminin_G_2	Laminin G domain	Domain	JBrowse
XP_048576726.1	PF00028	Cadherin	Cadherin domain	Domain	JBrowse
XP_048576726.1	PF01049	CADH_Y-type_LIR	Cadherin, Y-type LIR-motif	Family	JBrowse

Gene Family

Total gene families in 148 cnidaria organisms

Show: 10 /page genes per page

Gene family ID	Number of genes
OG0000000	25957
OG0000001	19508
OG0000002	22022
OG0000003	13835
OG0000004	12773
OG0000005	10756
OG0000006	11349
OG0000007	9044
OG0000008	10116
OG0000009	9470

This module shows a brief summary of clustered and annotated gene families.



Total members in OG0000799: 630

Show: 10 /page genes per page

Species	Gene ID	NR ID	Description	Visualization
Acropora acuminata	aacu_s0068.g57.t1	XP_015780410.1	XP_015780410.1 PREDICTED: protein bfr2-like [Acropora digitifera]	JBrowse
Acropora acuminata	aacu_s0079.g8.t1	CAH3184877.1	CAH3184877.1 unnamed protein product [Porites lobata]	JBrowse
Acropora acuminata	aacu_s0259.g7.t1	XP_044180355.1	XP_044180355.1 acidic leucine-rich nuclear phosphoprotein 32 family member B-like [Acropora millepora]	JBrowse
Acropora acuminata	aacu_s0412.g5.t1	PFX21924.1	PFX21924.1 hypothetical protein AWC38_SpisGene13592 [Stylophora pistillata]	JBrowse
Alatina alata	Aala_g1136.t1	CAB4007481.1	CAB4007481.1 PREDICTED: uncharacterized protein K02A2.6-like [Paramuricea clavata]	JBrowse
Alatina alata	Aala_g1156.t1	GFO48143.1	GFO48143.1 UDP-glucuronosyltransferase 2a1-like [Plakobranthus ocellatus]	JBrowse
Alatina alata	Aala_g1176.t1	CAH8540793.1	CAH8540793.1 unnamed protein product [Dicrocoelium dendriticum]	JBrowse
Alatina alata	Aala_g1207.t1	CAC5385438.1	CAC5385438.1 unnamed protein product [Mytilus coruscus]	JBrowse
Alatina alata	Aala_g1224.t1	CAC5385438.1	CAC5385438.1 unnamed protein product [Mytilus coruscus]	JBrowse
Alatina alata	Aala_g127.t1	MCO5550245.1	MCO5550245.1 hypothetical protein [Adiantum nelumboides]	JBrowse

This module shows the detailed information of family member in specific gene family.

Pan-geneset

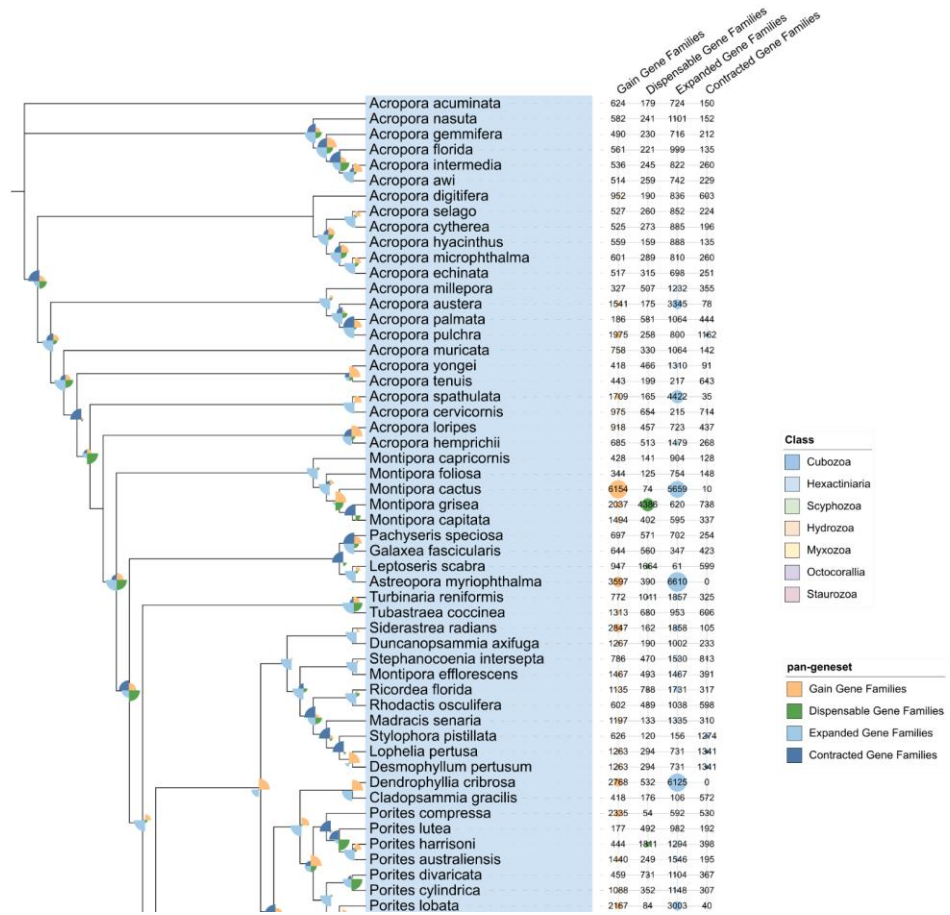
Pan-geneset

The information of Pan-geneset is identified using the [Count toolkit](#). The module also provides the gene family expansion and contraction information of each deep-sea species. The phylogenetic tree is constructed using the single-copy genes identified by OrthoFinder. Users can click the "Download" button to download the Pan-geneset data, including phylogenetic tree, gene family membership, and gene family expansion and contraction results.

Download

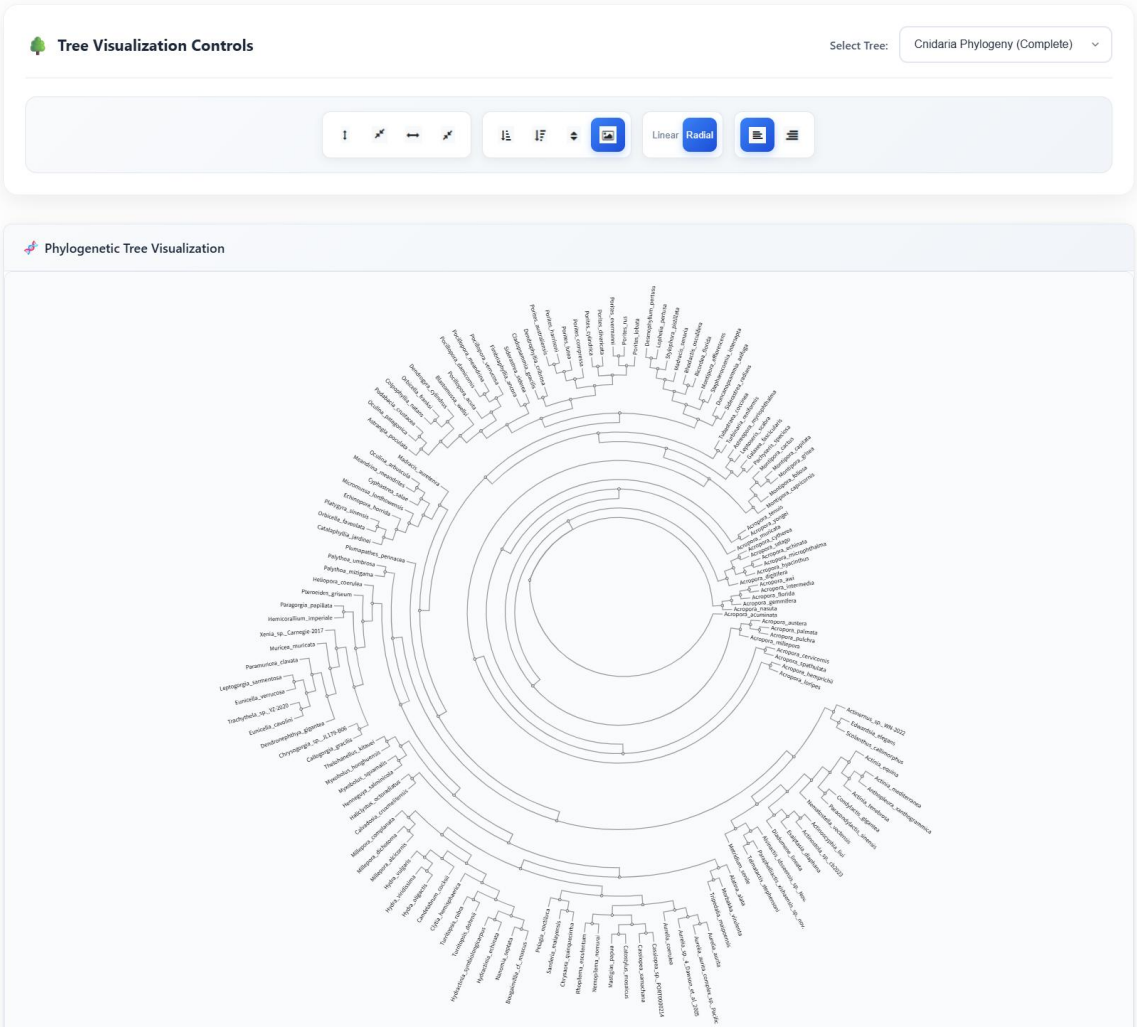


Users can download the Pan-geneset data, including phylogenetic tree, gene family membership, and gene family expansion and contraction results.



Pan-geneset data type

Species Tree



Transcriptomic Data

Transcriptomic Samples

A comprehensive collection of 15,642 transcriptomic samples across different adult tissues and developmental stages. Select a species to explore its transcriptomic data.

Class

Hexacorallia

Species

Acropora aff. hyacinthus sp. E JTL-2012

Choose class and species

View Samples



Detailed information of transcriptomic samples

Run	Class	Species	BioProject	SRA	Layout	Tissue	Developmental Stage	Treatment
SRX15278426	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	no treatment
SRX15278427	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	no treatment
SRX15278428	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch+ammonium salt+sediment
SRX15278429	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch+ammonium salt+sediment
SRX15278430	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	no treatment
SRX15278431	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch
SRX15278432	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch
SRX15278433	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch
SRX15278434	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch+ammonium salt
SRX15278435	Hexacorallia	Acropora microphthalma	PRJNA836701	SRP375259	PAIRED	skeletal tissue	-	to ^o Ch+ammonium salt

Co-expression Network

1. Select Targeted Species

Select Class

Hexactinaria

Select Species

Lophelia pertusa

2. Input Interested Gene List

Enter gene names, e.g., OS493_001533-T1 OS493_001609-T1 OS493_001606-T1
etc.

Example: "ABC transporter" [[Lophelia pertusa]]

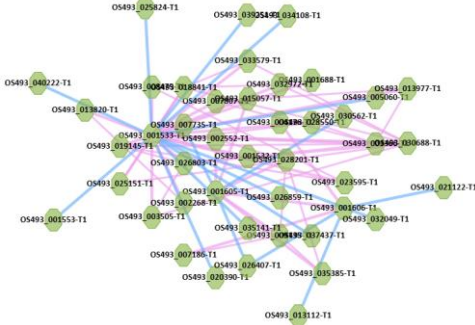
3. Select Co-expression Relationship

☒ Positive Co-expression ☒ Negative Co-expression

Note: "Positive" represents positive co-expression relationship with query gene. "Negative" represents negative co-expression relationship with query gene.

1. Choose species
2. Input your gene list like "ABC transporter related" list
3. Selected co-expression relationship

Co-expression network

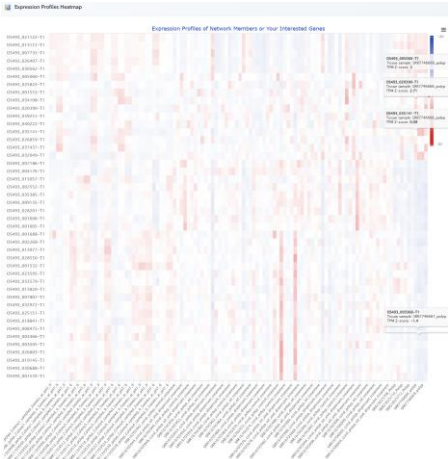


Detailed information of co-expressed genes

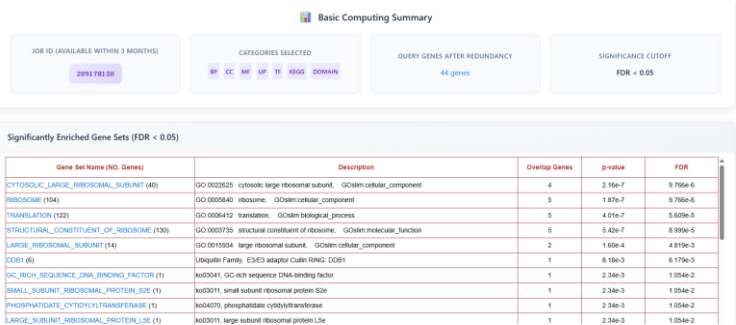
Co-expressed Genes of OS493_001533-T1 OS493_001609-T1 OS493_001606-T1

Gene ID	Description	PCC	Relationship
OS493_001533-T1	ATP-BINDING CASSETTE SUB-FAMILY C	1	positive
OS493_00066-T1	SCOTING NEKIN	0.85	positive
OS493_00140-T1	SHRMF-RELATED	0.83	positive
OS493_00063-T1	AT-RICH INTERACTIVE DOMAIN-CONTAINING PROTEIN 2	0.83	positive
OS493_00100-T1	LOXRNIP	0.82	positive
OS493_00100-T1	INICAL SPORE HEAD 1 HOMOLOG	0.82	positive
OS493_00045-T1	+	0.81	positive
OS493_00045-T1	CENTROSOMAL PROTEIN OF 250 KDA	0.81	positive
OS493_00101-T1	DNA REPAIR/RNA PROCESSING CSRF FAMILY	0.81	positive
OS493_00202-T1	CHROMOSOME ASSOCIATED KINESIN KIF4A-RELATED	0.80	positive
OS493_00107-T1	RIVEN CONA 402156656 GENE	0.80	positive
OS493_00100-T1	GC-RICH SEQUENCE DNA-BINDING FACTOR	0.80	positive
OS493_00107-T1	+	0.80	positive
OS493_00066-T1	SEMEY 26L7.2	0.80	positive

Expression profiling analysis



Gene sets enrichment analysis



Detailed information of co-expressed gene pairs used for dynamic network view

PROTEIN A	PROTEIN B	PCC	NR	RELATIONSHIP
OS493_001533-T1	OS493_001533-T1	0.7880000011480	16.105464214000	positive
OS493_001406-T1	OS493_001533-T1	0.81710000100204	1	positive
OS493_001406-T1	OS493_001533-T1	0.88821680000000	20.3714607016004	positive
OS493_001533-T1	OS493_001533-T1	0.77000000000000	20.3714607016004	positive
OS493_001533-T1	OS493_001533-T1	0.72124000000000	6.90207740000000	positive
OS493_001533-T1	OS493_001533-T1	0.82042000000000	6.24621010000000	positive
OS493_001415-T1	OS493_001533-T1	0.77000000000000	20.3707000000000	positive
OS493_001533-T1	OS493_001533-T1	0.82070000000000	6.30000000000000	positive
OS493_001533-T1	OS493_001533-T1	0.76000000000000	40.8410700000000	positive
OS493_001533-T1	OS493_001533-T1	0.78800000000000	4.38888888888889	positive
OS493_001415-T1	OS493_001533-T1	0.84870000000000	12	positive
OS493_001415-T1	OS493_001533-T1	0.78800000000000	47.8888888888889	positive
OS493_001533-T1	OS493_001533-T1	0.76770000000000	22.3000770000000	positive
OS493_001533-T1	OS493_001533-T1	0.78770000000000	22.3007700000000	positive
OS493_001533-T1	OS493_001533-T1	0.80070000000000	21.7704100000000	positive
OS493_001415-T1	OS493_001533-T1	0.84870000000000	9.18644000000000	positive
OS493_001415-T1	OS493_001533-T1	0.80070000000000	6.10447400000000	positive
OS493_001533-T1	OS493_001533-T1	0.78140000000000	28.3700219000000	positive
OS493_001533-T1	OS493_001533-T1	0.80070000000000	7.87070000000000	positive
OS493_001533-T1	OS493_001533-T1	0.84870000000000	21.0400000000000	positive
OS493_001533-T1	OS493_001533-T1	0.78140000000000	21.0400000000000	positive
OS493_001533-T1	OS493_001533-T1	0.78770000000000	14.4910700000000	positive

Positive represents positive co-expression relationship with query genes
Negative represents negative co-expression relationship with query genes

Dynamic Expression View

1. Select Target Species

Select Class

Hexactiniaria

Select Species

Lophelia pertusa

1. Choose species

2. Input Co-expressed Gene Pairs

OS493_001533-T1 OS493_001532-T1
OS493_001606-T1 OS493_001605-T1
OS493_001688-T1 OS493_001533-T1
OS493_002268-T1 OS493_001533-T1
OS493_002552-T1 OS493_001605-T1
OS493_003505-T1 OS493_001533-T1
OS493_004176-T1 OS493_001606-T1
OS493_003366-T1 OS493_001533-T1

Example: *Lophelia pertusa*

2. Input co-expressed gene pairs collected from network analysis

3. Input Genes with Expression Ratio

OS493_001533-T1 -0.577415084
OS493_001606-T1 -0.673612928
OS493_001688-T1 -0.800316114
OS493_002268-T1 -1.030611246
OS493_002552-T1 -0.611989434
OS493_003505-T1 -0.738763309
OS493_004176-T1 -0.604572254
OS493_003366-T1 1.031307454

Example: *Lophelia pertusa*

3. Input these genes and corresponding expression ratio

4. Select Threshold for Gene Expression Ratio

1 ≤ Red ≤ 100

-10 ≤ Blue ≤ -1

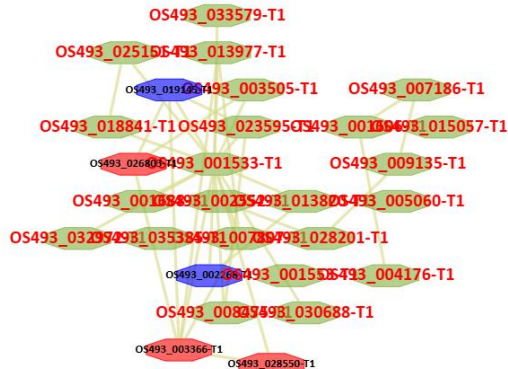
4. Select appropriate threshold for expression ratio

Note: Red indicates upregulated genes, blue indicates downregulated genes.

Submit Analysis

Reset Form

5. Dynamic network view



6. Expression profiling and gene sets enrichment analysis

Red represents upregulated genes in specific conditions
Blue represents downregulated genes in specific conditions

Cell Atlas

Select Tissue Sample

Species:

Acropora muricata

Stages:

Adult tissues/organs

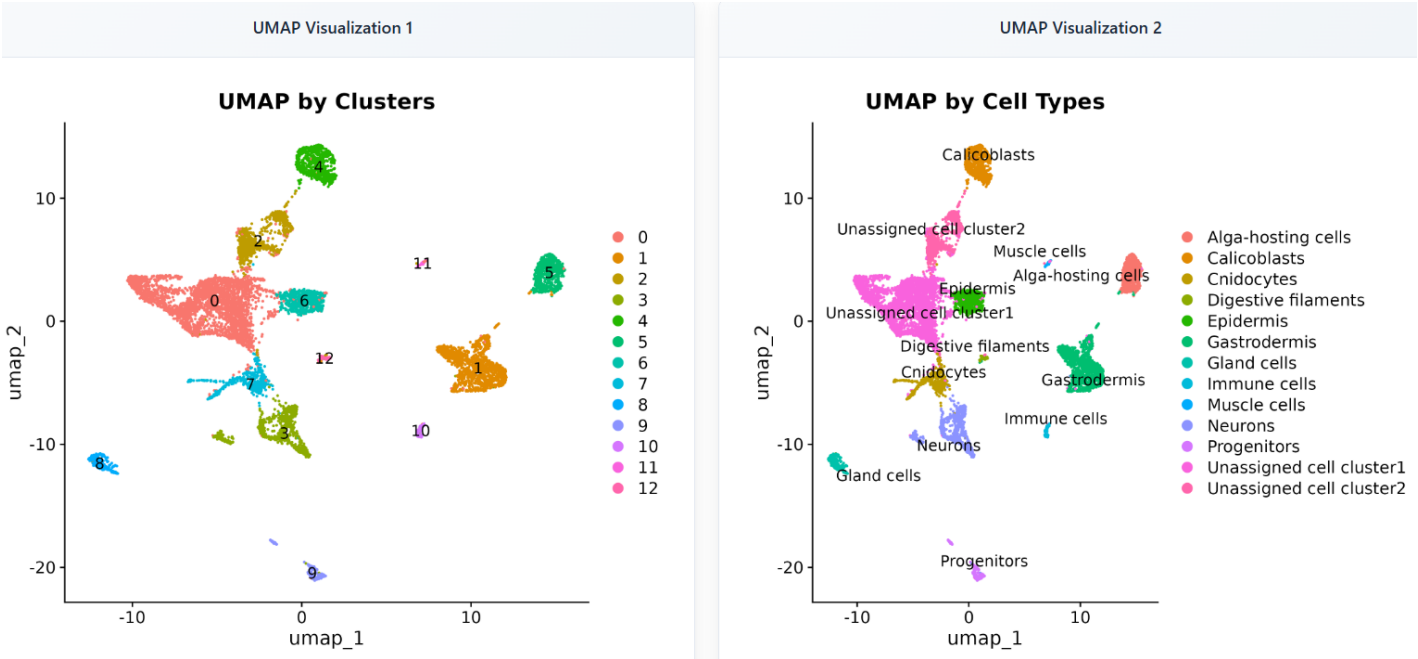
Emb/Org:

RegenerationStage

View Cell Atlas

Choose species, developmental stages and tissues

UMAP plot for different cell types



Cell Markers

Choose species, tissues, cell types and cluster id to view top 10 gene markers and detailed information of gene markers

Select Parameters for Cell Markers

Species

Acropora millepora

Stages/Org

Whole adults

Cell Type

calicoblast

Submit Query

Query Parameters

Species: Acropora millepora • Stage/Organ: Whole adults • Cell Type: calicoblast

4853 cell marker records found

- Top 10 Markers
- Amil_Amillepora19313

Amil_Amillepora10110

Amil_Amillepora03541

Amil_Amillepora19488

Amil_Amillepora12612

Amil_Amillepora37625

Amil_Amillepora32633

Amil_Amillepora14840

Amil_Amillepora28569

Amil_Amillepora15356

Top 100 cell Markers of calicoblast in Whole adults

Export to Excel

Cell Type	Gene	Symbol	log2FC	FDR	PC1	PC2
calicoblast	Amil_Amillepora19313		8.586101368	0	0.966	0.02
calicoblast	Amil_Amillepora10110		8.451916186	0	0.953	0.01
calicoblast	Amil_Amillepora03541	Dbh/Moxd1/Moxd2	7.368663415	0	0.968	0.034
calicoblast	Amil_Amillepora19488		7.472677481	0	0.856	0.018
calicoblast	Amil_Amillepora12612		8.323841742	0	0.833	0.014
calicoblast	Amil_Amillepora37625		5.82917066	0	0.844	0.039
calicoblast	Amil_Amillepora32633		3.721026458	0	0.827	0.07
calicoblast	Amil_Amillepora14840		9.342579347	0	0.758	0.002
calicoblast	Amil_Amillepora28569		2.671357191	0	0.919	0.169
calicoblast	Amil_Amillepora15356		5.029974002	0	0.771	0.031
calicoblast	Amil_Amillepora10108		8.284751895	0	0.737	0.004
calicoblast	Amil_Amillepora14146		8.286451043	0	0.739	0.006
calicoblast	Amil_Amillepora30757	Ncam1/Ncam2	3.578525789	0	0.863	0.142
calicoblast	Amil_Amillepora22520	Ltr/Meltr/Trf	4.494417219	0	0.785	0.07

Single-cell Gene Expression

Choose species, tissues, cell types and interested gene to view the gene expression patterns across different cell types

 **Query Gene Expression**

Species

Acropora muricata

Stages/Organs

Adult tissues/organs

Tissue/Organ

RegenerationStage

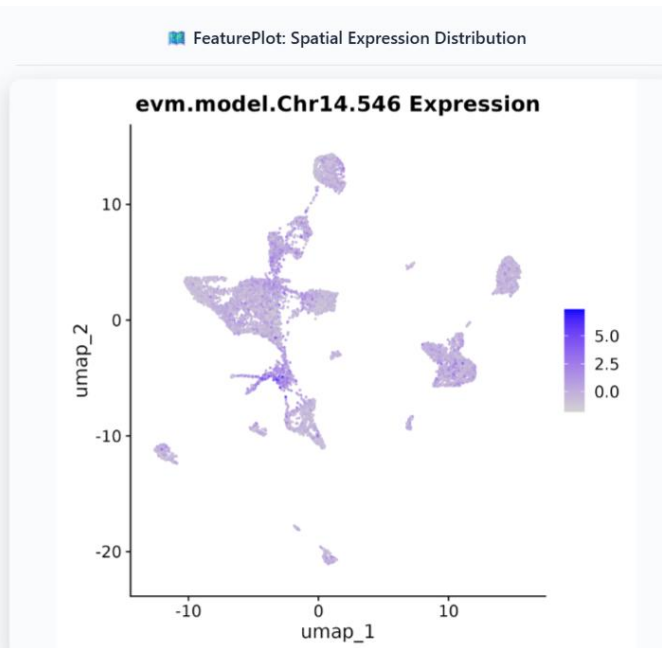
Gene ID

evm.model.Chr14.546

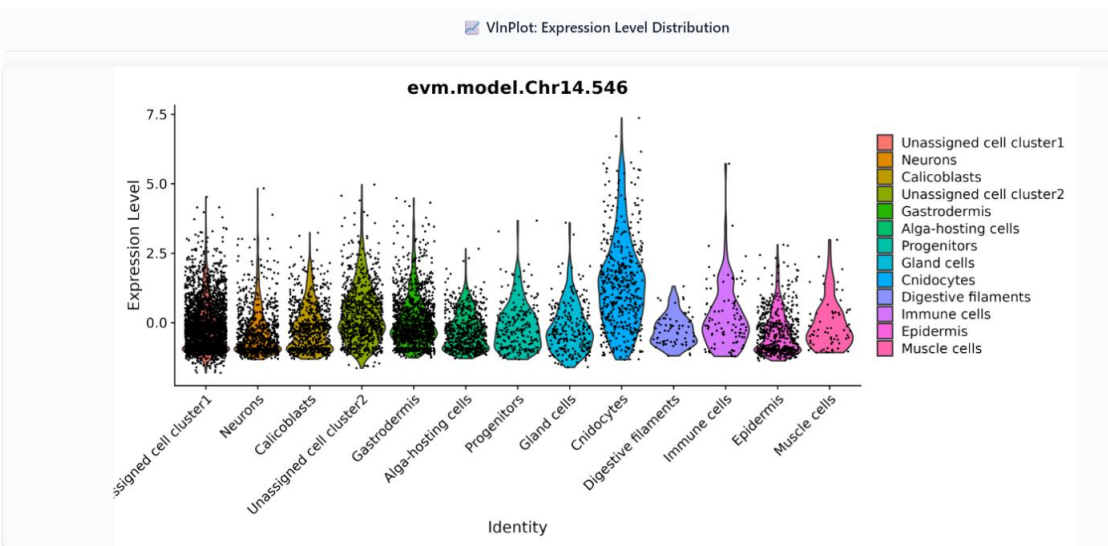
 **Submit Query**



UMAP plot for gene expression across cell types



Violin plot for gene expression across cell types



Proteomic Analysis

Species

Nematostella vectensis

Stages:

Adult tissues/organs

Emb/Org:

Embryos

View Samples

Search by protein, peptides, intensity...

Species	Tissue	Protein	q-value	Coverage [%]	Peptides	PSMs	Unique Peptides	MW [kDa]	calc. pI	Protein FDR Confidence	Modifications
Nematostella vectensis	Embryos	A7S177	0	56	84	1911	84	170.3	9.14	High	-
Nematostella vectensis	Embryos	A7S416	0	88	46	975	29	69.4	6.54	High	Met-loss [N-Term]
Nematostella vectensis	Embryos	A7RNL3	0	33	79	497	79	363.6	5.97	High	-
Nematostella vectensis	Embryos	A7RSB9	0	71	49	761	49	92.5	6.96	High	-
Nematostella vectensis	Embryos	A7RMC3	0	25	80	241	51	538.4	7.17	High	-
Nematostella vectensis	Embryos	P04264	0	55	41	737	34	66	8.12	High	-
Nematostella vectensis	Embryos	P35527	0	73	31	493	30	62.1	5.3	High	-
Nematostella vectensis	Embryos	A7SSW8	0	77	30	1152	29	51.7	8.97	High	-
Nematostella vectensis	Embryos	A7RM90	0	73	37	661	37	60.4	7.23	High	-
Nematostella vectensis	Embryos	P35908	0	70	40	421	31	65.8	8	High	Met-loss+Acetyl [N-Term]



Total Epigenomic Samples: 3768

Show:

10 /page

samples per page

Epigenomic Analysis

Species

Exaiptasia diaphana

Stages/Organs

Adult tissues/organs

Tissue/Organ

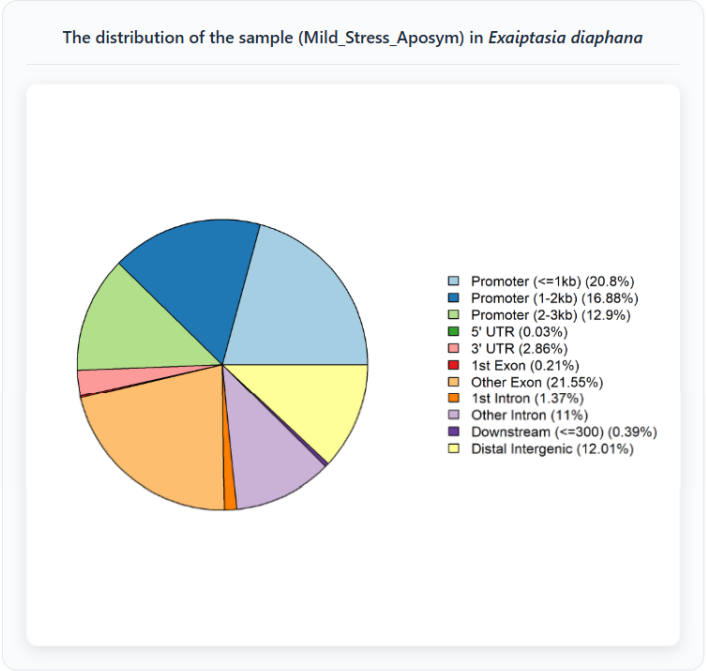
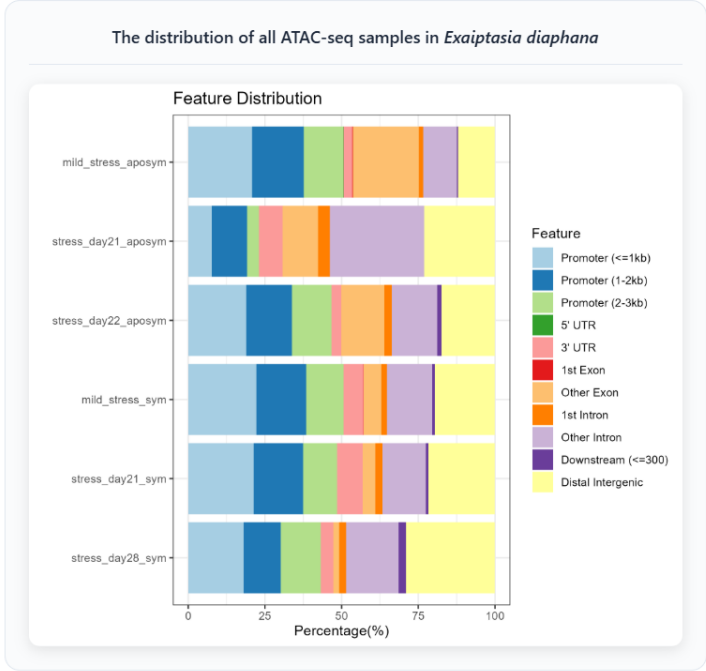
Mild_Stress_Aposym

Gene ID

KXJ30195.1

Submit Query

Visualization for the distribution of ATAC-seq in *Exaiptasia diaphana* - Mild_Stress_Aposym



Metagenome-Assembled Genomes

Class

Hexacorallia

Species

Acropora cytherea

View Samples

 **Query Parameters**

Class: Hexacorallia • **Species:** Acropora cytherea

 **14 metagenome-assembled genomes**

Search MAGs by Assembly Accession, Assembly Level..

Class	Host	Species	Assembly Accession	Assembly Level	Assembled Size (bp)	Nr.contigs	Contig N50 (bp)	GC Percent (%)	CheckM completeness (%)	CheckM contamination (%)	Paper
Hexacorallia	Acropora cytherea	Brevundimonas sp. A19_0	GCA_017744255.1	Contig	2922979	47	159010	67.5	98.03	1.85	41741644
Hexacorallia	Acropora cytherea	Endozoicomonas sp. G2_1	GCA_017744155.1	Contig	4125092	46	233877	42.5	99.95	1.75	41741644
Hexacorallia	Acropora cytherea	Labrenzia sp. R4_2	GCA_017743865.1	Contig	5660909	26	707340	56.5	94.89	8.09	41741644
Hexacorallia	Acropora cytherea	Labrenzia sp. R5_0	GCA_017743835.1	Contig	6776467	30	977995	59	96.16	9.07	41741644
Hexacorallia	Acropora cytherea	Paracoccus sp. R12_1	GCA_017744075.1	Contig	4068045	60	152621	64	99.34	1.41	41741644
Hexacorallia	Acropora cytherea	Qipengyuania flava	GCA_017744185.1	Contig	2951563	9	380715	63.5	96.54	2.79	41741644
Hexacorallia	Acropora cytherea	Ruegeria sp. R13_0	GCA_017743995.1	Contig	4687149	56	255467	58.5	98.2	4.91	41741644
Hexacorallia	Acropora cytherea	Ruegeria sp. R14_0	GCA_017743975.1	Contig	4659537	26	547728	58.5	98.59	4.32	41741644
Hexacorallia	Acropora cytherea	Ruegeria sp. R8_2	GCA_017743795.1	Contig	4185571	22	597609	57.5	98.89	3.39	41741644
Hexacorallia	Acropora cytherea	Salinisphaera sp. G21_0	GCA_017744115.1	Contig	6302811	312	92539	49	99.16	1.5	41741644

Phenotypic Information

Class	Species	Trait Name	Trait Type	Value	Trait Unit	Region	Latitude	Longitude	Methodology
Cubozoa	<i>Carybdea marsupialis</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Cubozoa	<i>Alatina alata</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Cubozoa	<i>Tripedalia cystophora</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Cubozoa	<i>Chiropsoides quadrigatus</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Cubozoa	<i>Chironex fleckeri</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Cubozoa	<i>Carybdea rastonii</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Cubozoa	<i>Chiropsalmus quadrumanus</i>	TG.carnivore	Ecological	1	-	-	-	-	converted from categorical
Hexacorallia	<i>Acanthastrea brevis</i>	Coloniality	Morphological	10	cat	Global estimate	-	-	Derived from textual description
Hexacorallia	<i>Acanthastrea brevis</i>	Growth form (typical hexacorallia)	Morphological	10	cat	Global estimate	-	-	Growth form derived from Veron (2000) textual descriptions
Hexacorallia	<i>Acanthastrea brevis</i>	Growth form Veron	Morphological	10	cat	Global estimate	-	-	Growth form derived from Veron (2000) textual descriptions



Total Taxon Information: 151046

Show: 10 /page records per page

Tools—Gene sets enrichment analysis

1. Select Target Species

Select Class **1. Choose species** Select Species

Hexactiniaria Lophelia pertusa

2. Choose Gene Sets for Analysis

2. Choose different types of background gene sets

☒ G1: Gene Ontology Gene Sets ☒ BP: Biological Process ☒ CC: Cellular Component ☒ MF: Molecular Function

☒ G2: Gene Family Based Gene Sets ☒ UBs: Ubiquitins ☒ TFs: Transcription Factors

☒ G3: KEGG Pathways ☒ G4: Protein Domain

3. Submit Your Query Gene List

Enter gene names (one per line)

 Load Example

OS493_001533-T1
OS493_001532-T1
OS493_001606-T1
OS493_001605-T1
OS493_001688-T1
OS493_002268-T1
OS493_002552-T1
OS493_003505-T1
OS493_004176-T1
OS493_003566-T1
OS493_007186-T1
OS493_007807-T1
OS493_008475-T1

3. Input different gene list such as DEGs list and co-expressed gene list



Detailed information of significantly enriched gene sets

Or upload a file (max 5MB)

 Choose File No file chosen

4. Choose Background Genes

☒ Suggested background (Whole genome level) ☐ Customized background

5. Parameter Selection (Optional)

4. Start analysis

 Start Analysis

 Reset Form

BLAST Program & Database

blastp (Protein-Protein)

Acropora acuminata pep

Query Sequence

Enter sequence in **FASTA** format

7x090041.cts

Or upload a FASTA file (max 5MB)

Choose File No file chosen

 Search Parameters

 Expect Threshold

10 (Default)

Scoring Matrix

BLOSUM62 (Default)

 Alignment Options

☐ Perform ungapped alignment☒ Include graphical overview Results Display

Descriptions

100 (Default)

Alignments

50 (Default)

 [Start BLAST Search](#)

[Reset Form](#)

2. Input your sequences in fasta style

Blast results

Query ID	Subject ID	Alignment Length	Mis Matches	Query Start	Query End	Subject Start	Subject End	% Identity	E-value	Bit Score
s0340.g1.t1	v0404.g1.t1	939	0	1	939	1	939	100.000%	0.0	1960
s0340.g1.t1	v0404.g2.t1	941	121	1	939	1	937	86.504%	0.0	1682
s0340.g1.t1	v0404.g2.t2	940	195	1	939	1	888	73.617%	0.0	1408
s0340.g1.t1	v0340.g1.t1	888	165	53	939	10	845	75.450%	0.0	1355
s0340.g1.t1	v0404.g1.t1	995	39	545	939	1	395	90.127%	0.0	739
s0340.g1.t1	v0404.g1.t1	915	419	1	879	1	879	43.370%	0.0	712
s0340.g1.t1	v0404.g1.t1	906	889	1	936	176	1158	39.124%	0.0	651
s0340.g1.t1	v0505.g16.t1	295	119	640	934	1	299	57.960%	7.70e-115	303
s0340.g1.t1	v0505.g16.t2	323	117	640	934	1	319	53.870%	2.86e-111	345
s0340.g1.t1	v0404.g3.t1	215	41	53	265	377	591	60.000%	6.67e-111	364
s0340.g1.t1	v0404.g3.t1	343	202	517	838	608	943	32.945%	4.65e-100	191
s0340.g1.t1	v0258.g19.t1	601	349	263	838	279	862	34.942%	3.59e-103	342
s0340.g1.t1	v0404.g3.t1	186	30	496	679	112	297	62.791%	4.47e-91	296
s0340.g1.t1	v0404.g3.t1	140	21	800	939	297	436	85.000%	9.88e-75	252
s0340.g1.t1	v0404.g3.t1	111	33	168	278	1	111	70.270%	2.38e-47	175
s0340.g1.t1	v0206.g1.t1	272	150	582	853	240	503	41.912%	1.41e-63	223
s0340.g1.t1	v0066.g05.t1	272	142	583	851	199	454	42.222%	2.77e-62	218
s0340.g1.t1	v1442.g1.t1	269	150	582	848	116	374	39.777%	7.65e-60	209
s0340.g1.t1	v0258.g16.t1	269	153	582	849	372	630	39.050%	1.52e-59	215
s0340.g1.t1	v0258.g16.t1	179	83	493	679	1	85	39.050%	1.92e-59	211

Tools—Primer design

Primer3Plus

pick primers from a DNA sequence

More...

Help

Source Code

About

Load server settings: Default

Activate Settings

Task: generic

Select primer pairs to detect the given template sequence. Optionally targets and included/excluded regions can be specified.

Pick Primers

2

MainGeneral SettingsAdvanced SettingsInternal OligoPenaltiesAdvanced Seq.

Sequence Id: Medicago_Lyk3

Paste template sequence below or upload sequence file: 选择文件 未选择任何文件 Load Example

1

acaatatgttattgttgagatcatataagattgtagtcacacattcttgtaaaaggtctcagattcattcccccgtatacaatttaagt
gagctaatttagcttcttaaaaaataaaacaaacatttacaataaactcagtgaaacttggaataaagtatctctatactactct
ttagcttgattgattctctcgaagaattattatagctttgattatcatcaagaacattcattgaattttaagtaagaattacta
acattgcaactcttattctagcaacagttactttgtccctccacattctcttgaastagtcacaaagtcacacatcatgcatctgt
ctgatacatcaacgaagaagcatgtagaagaattgtagtgaaagtagagtagcatttccctcaaaaaattatcaaaagtgacat
tatctttctgtagaattaaacattttttgctgatttgagattct
ttct
ttgaatccaaatgtgtaaaaggtgtgtagcttttagcttctactatattaccatcaactcaactcagaatatatcaaacattta
tgcaatcaagaattgtttaccacattctcttgatttataatgagctacaatagagacgtagtagtcaaatctctgtcttattct
atactagaatcaacgttccgttcccatgtgtaagtagtggaagtggaattctagacattgttgaattgaataacacaaagaagagagac
attatgatttaattgcaaatattatagcgaagtttgacaacgttgtagttatgaaaaagtcacacagctatgacccaatcatatc
ctgttaagctaaagattaaagtcacgttaattgttcattgggaatagccagatttcaaaagattatgctgtttgttgaactatcac

Mark selected region: < Excluded > [Target] [Included] Clear Regions from Seq. Save Sequence

Excluded Regions: < [] >

Targets: [[]]

Included Region: { [] }

Primer overlap positions: - []

Pair OK Region List: []

☒ Pick left primer

☐ Pick hybridization probe

☒ Pick right primer

or use left primer below.

(internal oligo) or use oligo below.

or use right primer below (5'->3' on opposite strand).

[]

[]

[]

5' Overhang: []

5' Overhang: []



MainGeneral SettingsAdvanced SettingsInternal OligoPenaltiesAdvanced Seq.

Results

☐ Select all Primers

Send Selected Primers to Primer3Manager

Go to Primer3Manager

First Primer PairPrevious Primer PairNext Primer PairLast Primer Pair

☒ Pair 1: Medicago_Lyk3

Left Primer 1: ggtcaatgcttcctgtgagc

Start: 1881 Length: 20 bp Tm: 59.2 C B: 93.295 % GC: 55.0 % Any: 0.0 End: 0.0 TB: 7.0 HP: 40.5 3' Stab: 4.3 Penalty: 0.807

Hairpin Secondary Structure:
Tm: 40.5° C dG: -294 cal/mol dH: -26100 cal/mol dS: -83 cal/mol*K
5' GGTCATGCTTCCTG
||| T
3' CAGATG

Right Primer 1: cctgcctctaagtcttgcga

Start: 2475 Length: 20 bp Tm: 59.5 C B: 93.744 % GC: 55.0 % Any: 0.0 End: 0.0 TB: 9.0 HP: 0.0 3' Stab: 4.4 Penalty: 0.533

Pair: Product Size: 595 bp Tm: 81.5 C Any: 0.0 End: 0.0 TB: 16.0 Penalty: 1.340

First Primer PairPrevious Primer PairNext Primer PairLast Primer Pair

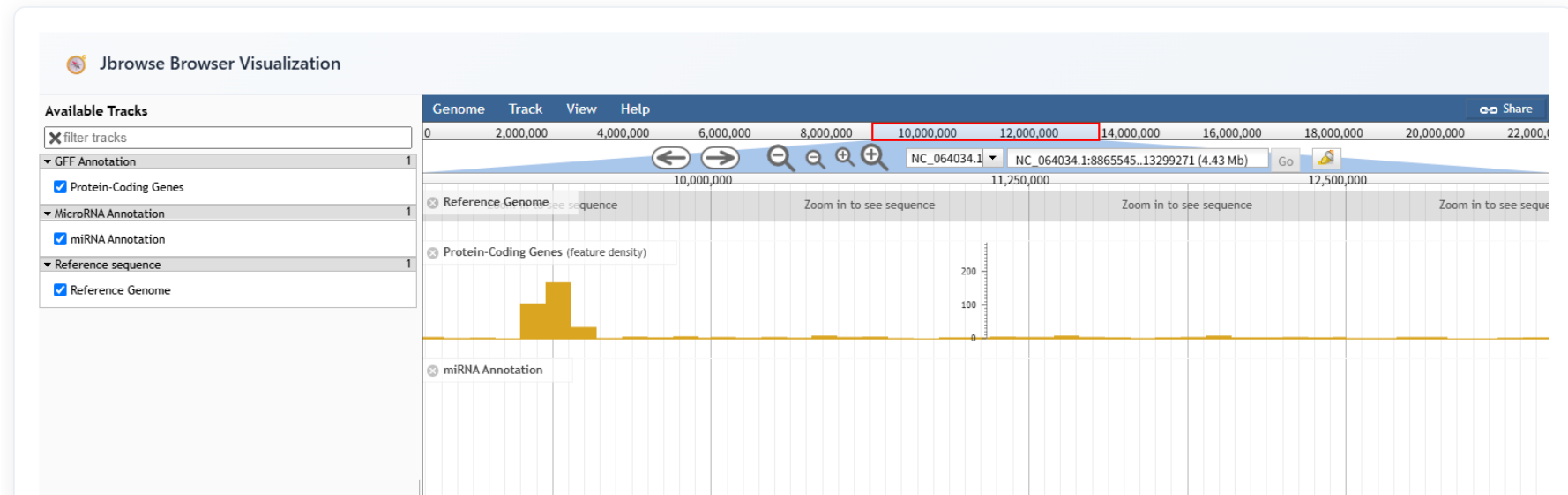
1	acaatatgtt	attggtgaga	tcatataaga	tttgatgtca	acatcttcgt
51	aaaggtctca	gattcgattc	tccccgggat	caatttaagt	gagctaattt
101	agcttcttaa	aaaaataaat	caaacacatt	ttacataaac	tcagtgaaaa
151	cttggaata	aagtatcctt	atactactct	ttagtcttga	ttagtctctg
201	caagatatt	tatatgtact	ttgtattatc	ataagaacat	tcattgacat
251	ttaagattaa	tgaattacta	acatgtcaac	tcttattcta	gccaacagtt
301	actttgttcc	ctccacattc	tctttgaaat	agtcacaaag	atccaatcat
351	gcattgtctg	tctgatcata	acagcaaaa	catgtgtata	gaaaaattgat
401	agtggaatta	gagtcatttt	ccataaaaa	atattcaata	agtgtagcat
451	tatttttctg	atgaattaat	ccattttttg	ctgatttgag	attctttctt
501	tctttgtctc	ttgttttctc	tcatacgcca	ttttttttgt	ttctctcttc
551	ctctctctct	cttgattcaa	tgaattctca	aaatgatta	ctattgttca
601	tctgtttctc	ggattgtgtt	tttttcaaa	ttgaatccaa	atgtgttaaa
651	gggtgtgatg	tagcttttag	ttctactat	attataccat	caattcaact
701	cagaaatata	tcaaatctta	tgcaatcaaa	gattgttctt	accaattctc
751	ttgatgttat	aatgagctac	aatagagaag	tagtattoga	taaatctggt

Tools—Jbrowse visualization

 **Select Species for Visualization**

Species

Nematostella vectensis



Download

In this module, users can click the "Download" button to download genome and gene sequences in FASTA format, gene annotations in GFF3 format, and functional annotation including interpro, protein domain, GO terms, KEGG pathway and gene family.

[illegible]

Statistics

Database Content Statistics

Class/Order/Family/Species	7/22/107/325
Assembled Genomes With Full Annotation	145
Tracks in iBrowse	3,141
Paleobiological Records	410
Busco genes	542,399
Protein-Coding Genes	5,450,036
Mitogenomic Datasets	78
Transcriptomic Datasets	15,642
Sing-cell RNA-seq Datasets	33
Proteomic Datasets	46
Epigenomic Datasets	723
Metagenomic Datasets	1,650
Phenotype Datasets	151,046
Co-Expressed Networks/Gene Pairs	31/13,793,211
Cell types	206
Cell markers	297,645
Quantified Proteins	63,306
Epigenetic Modification Type/Peaks	13/79,820,475
Metagenome-Assembled Genomes	315
NR/Uniprot/GO/KEGG/Plam/Panther/Interpro annotation	69,363,242/140,368,972/96,073,030/1,362,960/7,479,863/3,805,615/55,694,300
TPs/Ubiquitin families	96,654/206,659
Orthologous Families	3,621,064

Summary of data composition

Software and Analytical Tools

Module	Software	Parameter	Links
Fossil records	PSOB	default	https://paleobiodb.org/
Gene annotation	InterProScan v.5.67.67.0	-f tsv -g terms -pa	https://github.com/ebi-pf-team/interproscan
Busco Gene	BUSCO v.5.8.2	-i metatool -od12 -m proteins	https://github.com/estabib/busco
Transposable Elements	RepeatModeler v.2.0.7	-engine mbias -L TRSstruct	https://github.com/CTAM-consortium/RepeatModeler
	RepeatMasker v.4.2.1	-gff -e mbias -nrow	https://github.com/CTAM-consortium/RepeatMasker
TFS/Ubs	HMMER v.3.4	cut-off: 0.0001	http://hmmer.org/
Gene Family	OrthoFinder v.2.5.5	default	https://github.com/davidemms/OrthoFinder
Pan-genaset	Count	Wagner_parsimony -gain 1 -max_paralogs 200 Posterior -max_paralogs 200	https://www.iro.umontreal.ca/~curois/gene_content/count.html
Mitogenomic Datasets	OQORAW v.1.3.1	default	https://ichlorobox.mpimp-goim.mpg.de/OQORaw.html
Proteomic Datasets	blastp	default	https://blast.ncbi.nlm.nih.gov/Blast.cgi
Epigenomic Datasets	MACS3	default	https://github.com/macs3-project/MACS
Co-expression network	Cytoscape v3.10.3	default	https://cytoscape.org/
Single-cell Datasets	Seurat v4.4	LogNormalize	https://satijalab.org/seurat
Species tree	phylotree	default	https://www.phylotree.org/
Primer Design	Primer3Plus	default	https://github.com/primer3-org
Database construction	LAMP	default	https://github.com/teddyxun/lamp

Softwares used for data analysis

Data Submit

Name *

Jackie

Position *

Faculty

Organization *

HKUST

Email *

yourEmail@gmail.com

Species *

Gigantidas platifrons

Data Type *

Genome (fasta and GFF3, etc.)

Data Upload Link *

Provide data links, separated by commas

Data Description

Brief description of the uploaded data (optional)

Submit Data

Reset Form

Please don't hesitate to leave your comments here if you have any suggestions or comments.

Community Feedback

1 Comment

Jiajie She

J

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Share

BestNewestOldest

J

Jiajie She

8 days ago

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Thank You!